

FROM SYMBOLS TO MICROSECONDS: FUSION-PLASMA NUMBERS FOR THE CONE’S BATH THEOREMS

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ABSTRACT. [1] derives a diffusion coefficient $\text{Tr } D$ and a friction coefficient A_{friction} for a charged test particle in a plasma directly from binary Rutherford encounters, verified only against its own Monte Carlo simulation. This note asks whether that derivation, with zero fitting to real data, produces a sensible number for a real physical system. Taking ITER-core-like parameters ($n_e = 10^{20} \text{ m}^{-3}$, $\Theta_e = 10 \text{ keV}$) and computing every intermediate quantity — Debye length, $\ln \Lambda$, $\text{Tr } D$, A_{friction} — from first principles gives a collision time of $23.6 \mu\text{s}$. Comparing against two independently-sourced, properly cited standard results (Braginskii’s electron-electron collision time and the Spitzer–Landau formula, both an order of magnitude larger at $122\text{--}178 \mu\text{s}$), we do not stop at “same order of magnitude”: we trace the gap to its source. Our formula scales as $\tau \propto v_{\text{th}}^3$; using the root-mean-square thermal speed $v_{\text{rms}} = \sqrt{3\Theta/m}$ implicit in the Spitzer–Landau derivation, rather than the single characteristic speed $\sqrt{\Theta/m}$ we used initially, reproduces the Spitzer–Landau number to 5.4% — within our own previously-flagged $O(1/\ln \Lambda) \approx 25\%$ leading-log accuracy. A residual factor of 1.5 against Braginskii’s more carefully Maxwellian-averaged formula remains, and we attribute it honestly to the difference between evaluating our coefficients at one characteristic speed and performing the full velocity-space average his derivation carries out, which we have not done. We also flag, rather than paper over, that $v_{\text{th}} \approx 0.14c$ at these parameters, mildly outside the strict validity of the nonrelativistic derivation throughout.

1. THE QUESTION

Every theorem in [1] was derived and verified in isolation — against symbolic algebra, against a purpose-built Monte Carlo simulation, against internal consistency checks. None of it was checked against an independent, real physical number. This note does that once, for the most directly testable of the three baths considered there: a real plasma.

2. THE SYSTEM

Electron–electron Coulomb collisions in an ITER-core-like tokamak plasma:

$$n_e = 10^{20} \text{ m}^{-3}, \quad \Theta_e = 10 \text{ keV}.$$

These are standard, unremarkable fusion-plasma-core parameters, chosen before any calculation for being typical rather than for producing a good match.

3. OUR OWN PREDICTION

Using only the formulas of [1] and no external input beyond the parameters above and fundamental constants:

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$v_{\text{th}} = \sqrt{\Theta_e/m_e}$	$4.19 \times 10^7 \text{ m/s } (0.140c)$
Debye length λ_D	$74.3 \text{ } \mu\text{m}$
$b_{90} = \kappa_{\text{int}}/(\mu_{\text{red}} v_{\text{th}}^2)$	$2.88 \times 10^{-13} \text{ m}$
$\ln \Lambda = \ln(\lambda_D/b_{90})$	19.4
$\text{Tr } D$ [eq. (4) of [1]]	$7.45 \times 10^{19} \text{ m}^2/\text{s}^3$
A_{friction} [eq. (5)]	$-1.78 \times 10^{12} \text{ m/s}^2$
$\tau := v_{\text{th}}^2/\text{Tr } D = v_{\text{th}}/ A_{\text{friction}} $ [identical, eq. (6)]	$23.6 \text{ } \mu\text{s}$

The two routes to τ agree exactly, as they must given the exact ratio (6) already proved in [1]; this is a consistency check on our own bookkeeping, not an independent validation.

4. COMPARISON TO ESTABLISHED PLASMA PHYSICS

Proposition 4.1 (Two independent literature formulas). *The Braginskii (1965) electron-electron collision time [2, 3] and the Spitzer–Landau formula [4] are, respectively,*

$$(8) \quad \tau_{ee}^{\text{Brag}} = \frac{6\sqrt{2}\pi^{3/2}\epsilon_0^2\sqrt{m_e}\Theta_e^{3/2}}{e^4 n_e \ln \Lambda}, \quad \tau_{ee}^{\text{SL}} = \frac{2\pi\epsilon_0^2\sqrt{m_e}(3\Theta_e)^{3/2}}{n_e e^4 \ln \Lambda}.$$

Numerical Evaluation. Using our own $\ln \Lambda = 19.4$ for a direct, apples-to-apples comparison:

Formula	τ_{ee}
Ours (Section 3, $v_{\text{th}} = \sqrt{\Theta/m}$)	$23.6 \text{ } \mu\text{s}$
Braginskii (8)	$177.6 \text{ } \mu\text{s}$
Spitzer–Landau (8)	$122.7 \text{ } \mu\text{s}$

Same order of magnitude, but a genuine factor of 5–8 discrepancy — larger than the $O(1/\ln \Lambda) \approx 25\%$ accuracy we would expect from the leading-log approximation alone, and worth chasing down rather than waving at.

Proposition 4.2 (The discrepancy is a velocity-convention effect, almost entirely). *Our τ scales as v_{th}^3 (since $\text{Tr } D \propto 1/v_{\text{th}}$ and $\tau = v_{\text{th}}^2/\text{Tr } D$). The ratio to Spitzer–Landau implies an effective velocity $1.733\times$ ours; to Braginskii, $1.960\times$ ours. Recomputing τ with $v_{\text{rms}} = \sqrt{3\Theta_e/m_e}$ — the root-mean-square speed, matching the explicit $(3\Theta_e)^{3/2}$ appearing in the Spitzer–Landau formula (8) — in place of $v_{\text{th}} = \sqrt{\Theta_e/m_e}$ gives $\tau = 116.2 \text{ } \mu\text{s}$.*

Numerical Verification. $|116.2 - 122.7|/122.7 = 5.4\%$ against Spitzer–Landau: essentially exact agreement once the same characteristic speed is used, well inside our own flagged leading-log accuracy. Against Braginskii, a residual factor of $177.6/116.2 = 1.53$ remains.

Remark (What the residual factor is, honestly). We do not close this remaining gap. Braginskii’s formula is derived by properly averaging the Rosenbluth-potential Fokker-Planck coefficients over the full Maxwellian distribution, not by evaluating a single-speed formula at one characteristic velocity; that averaging introduces genuine numerical factors (the $\sqrt{2}\pi^{3/2}$ structure visible in (8)) beyond a simple choice of v . Our derivation in [1] used an idealized monoenergetic field, precisely so that the leading-log calculation would be transparent and checkable by direct Monte Carlo; doing the analogous Maxwellian average honestly is a further calculation we have not carried out, and $1.5\times$ is a plausible, unsurprising size for what it would change.

5. THE CAVEAT WE ARE NOT PAPERING OVER

$v_{\text{th}} = 0.140c$ at $\Theta_e = 10$ keV. Every formula used here — Rutherford scattering, elastic two-body kinematics, the Fokker-Planck coefficients of [1] — is nonrelativistic. Corrections at $v/c \sim 0.14$ are formally $O(v^2/c^2) \sim 2\%$, likely smaller than the velocity-convention and Maxwellian-averaging effects already identified above, but we have not computed them, and a fully honest application of this machinery to a specific real device would need to.

6. DISCUSSION

This is a modest but genuine validation. The formulas of [1] were derived from first principles and checked only against a Monte Carlo simulation of their own idealized model; they were never fit to, or informed by, real plasma data. Applied cold to unremarkable ITER-core parameters, they produce a number within 5.4% of an independently-derived, properly cited textbook result, once one accounts for a specific, identified, and quantitatively confirmed difference in convention (which characteristic speed to use) rather than an unexplained fudge factor. That is a meaningfully stronger statement than “same order of magnitude”: it says the functional form we derived — $\text{Tr } D \propto n\kappa_{\text{int}}^2 \ln \Lambda / (m^2 v)$ and its friction counterpart — is not merely dimensionally sensible but quantitatively correct, up to exactly the kind of $O(1)$ conventions that any leading-log Coulomb-scattering calculation carries. It is not evidence that the broader “effective field theory for slow orbital variables” framing motivating this note is a rigorous EFT in the technical sense; it is evidence that the specific coarse-graining calculation underneath it captures real physics rather than being a self-consistent but empty formal exercise.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic’s Claude) were used in the derivation, literature search, numerical computation, and L^AT_EX preparation of this note. This includes: numerical evaluation of Section 3’s derived quantities from fundamental constants and the stated plasma parameters; a web search to locate properly citable, independent formulas for the standard electron-electron collision time (Proposition 4.1), replacing an initial comparison against a formula recalled from memory with uncertain precision; numerical evaluation of both literature formulas; and the diagnostic calculation in Proposition 4.2 tracing the discrepancy to the choice of characteristic velocity, verified by direct recomputation with v_{rms} substituted for v_{th} . All mathematical and numerical claims were independently verified by the author.

REFERENCES

- [1] J. Ebert, *The Cone in a Bath: Boson, Gravitational, and Plasma Relaxation of the Semi-Major Axis*, Preprint (2026).
- [2] S. I. Braginskii, Transport processes in a plasma, in *Reviews of Plasma Physics*, Vol. 1, Consultants Bureau, New York, 1965.
- [3] Coulomb collisions in strongly anisotropic plasmas I: cyclotron cooling in electron-ion plasmas, *arXiv:2006.09246*, eq. (2.15) (statement of the Braginskii electron-electron/electron-ion collision time).
- [4] The role of collisions and strong coupling in ultracold plasmas, *arXiv:1201.0789*, eq. (2) (statement of the Spitzer–Landau collision time).

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